

Formulation Guidelines for Sytenol® A

Sytenol® A is an all natural and well-defined material with a purity level of $\geq 90\%$. Formulated product containing Sytenol® A will be white and remain white in appearance in formulated products. At 45 °C for three months, a lotion containing 1% Sytenol® A showed practically no color shift.

We have developed a series of application guideline tips that will assist the formulator to make products that are as aesthetically pleasing and stable as possible.

- ◆ Sytenol® A should be used at a level of 0.5 – 1.0% (w/w) of finished formulation.
- ◆ Sytenol® A is miscible in a wide variety of emollients (for example, capric/caprylic triglycerides, C₁₂₋₁₅ alkyl benzoates, and mineral oils), vegetable oils (sunflower oil, jojoba oils, olive oil) and silicones (dimethicones, cyclomethicones).
- ◆ Dissolve Sytenol® A in appropriate hydrophobic emollients and add to the formulation after emulsification. Solution of Sytenol® A may be added to the formulation with a processing temperature of about 50°C or below. Alternately, Sytenol® A can be added directly to the oil phase. Avoid prolonged heating of the oil phase above 75 °C.
- ◆ For preparation of anhydrous serum or transparent gel, use silicone derivatives such as, Dimethicone, Cyclomethicone, and Dimethiconol & Dimethicone Crosspolymer.
- ◆ Sytenol® A is a phenolic compound. Therefore, presence of iron or copper ions must be avoided to eliminate coloration due to the interaction with these metal ions. Addition of a small amount of disodium EDTA (0.1%) or other chelating agents resolves this problem.
- ◆ The finished product must be acidic, preferably having pH below 6.5. A pH below 5.5 is recommended when potassium sorbate or sodium benzoate are used as preservatives.
- ◆ Avoid prolong exposure to sunlight or heat. Opaque and airless packaging is highly recommended for maintaining product integrity and stability over time.

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Anti-Aging Night Cream with 0.5% Sytenol® A

Formulation#RC-01-13

INCI name	Trade Name/Supplier	% w/w
Phase A1		
Water	Water(demineralized)	Qs
Disodium EDTA	Versene Na/Dow	0.10
Glycerin	Emery 916/Cognis	3.00
Butylene Glycol	Butylene Glycol/Ruger	2.00
D Panthenol	Panthenol/BASF	0.50
Phase A2		
Xanthan Gum	Keltrol® CG-T/CP Kelco	0.15
Carbomer	Carbopol Ultrez 10/Lubrizol	0.40
Sucrose Stearate	Systema SP50C /MMP	2.50
Poly C8-22 Alkyl Acrylate/Metacrylic Acid Cross Polymer	Intelimer 8600/Air Products	2.50
Phase B		
Simmondsia Chinensis (Jojoba Oil)	Jojoba Oil/Jeen Intl.	3.00
Shea Butter	Shebu Refined / Rita	2.50
Behenyl Alcohol	Lanette 22/BASF	1.50
Potassium Cetyl Phosphate	Amphisol® K /DSM	1.00
Stearic Acid	Pristerene® 4911/Croda	1.00
Dimethicone	Dow Corning 200, 100cst/Dow Corning	4.00
Natural Beeswax	Beeswax/Rita	2.00
Isosorbide Dicaprylate	Synovea® DOI/Sytheon	3.00
Alpha Tocopherol	Vitamin E/Basf	0.20
Hydrogenated Polyisobutene	Panalane L14E/Lipo	4.00
Bakuchiol	Sytenol® A /Sytheon	0.50
Phase C		
Sodium Hydroxide	Sodium Hydroxide (50%sol)	0.40
Phase D		
Fragrance	Women's FragranceK3258/Carruba	0.20
Red 33 (CI 17200)	D&C Red 33 (1% sol)	0.05
Phenoxyethanol, Ethylhexylglycerine	Euxyl PE 9010/Schulke	1.00
Total		100.00

Procedure:

Combine A-1; disperse A2 one by one in A1 while stirring and heat A to 75°C. Combine ingredients of phase B and heat to 75 °C. Add phase B to A with good mixing. Homogenize mixture at moderate speed for 3-5 min while adding phase C (adjust pH to 5.5-6.0). Cool batch to 40°C with propeller agitation until mixture is homogeneous. Add phase D ingredients while continue mixing.

Notes:

pH - 5.5-6.0 Viscosity – 80,000- 100,000mPas (Brookfield RVT, Spindle C, 10 rpm) at 25°C

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Anti-Aging Repair Serum with 0.5% Sytenol® A

Formulation#RC-01-34

INCI name	Trade Name/Supplier	% w/w
Phase A1		
Water	Water(demineralized)	Qs
Disodium EDTA	Versene Na/Dow	0.10
Glycerin	Emery 916/Cognis	3.00
Butylene Glycol	Butylene Glycol/Ruger	2.00
D Panthenol	Panthenol/BASF	0.20
Phase A2		
Acrylates/C10-30 Alkyl Acrylate Crosspolymer	Carbopol Ultrez 21/Lubrizol	0.15
Phase B		
Triethanolamine	Triethanolamine 99%/Ruger	0.10
Phase C		
Isosorbide Dicaprylate	Synovea® DOI/Sytheon	2.00
Isostearyl Alcohol, Butylene Glycol Cocoate, Ethylcellulose	Emulfree CBG/Gattefosse	2.00
Cyclopentasiloxane, Dimethiconol	Dow Corning 1501 Fluid/Dow Corning	2.50
Bakuchiol	Sytenol® A/Sytheon	0.50
Phase D		
Hydroxyethylacrylate/sodium Acryloyldimethyltaurate Copolymer, Squalane, Polysorbate 60	Simulgel NS/Seppic	1.50
Phase E		
Poly C8-22 Alkyl Acrylate/Metacrylic Acid Cross Polymer	Intelimer 8600/Air Products	2.00
Phase F		
Fragrance	Green Citrus#61044049/Bell Flavors	0.10
Phase G		
Phenoxyethanol, Ethylhexylglycerine	Euxyl PE 9010/Schulke	1.00
Total		100.00

Procedure:

Combine A-1; disperse A2 one by one in A1 while stirring and heat A to 40°C. Combine ingredients of phase C and heat to 40 °C. Neutralize phase A with phase B to pH 5.5. Add phase C to AB with good mixing. Homogenize mixture at moderate speed for 3-5 min while adding phase D. Switch batch to propeller agitation until mixture is homogeneous. Add phase E, F & G while continue mixing.

Notes:

pH - 5.5-6.0 Viscosity – 25,000- 35,000mPas (Brookfield RVT, Spindle C, 10 rpm) at 25°C

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Age-Defying Day Cream with 1% Sytenol® A

Formulation#RC-01-42

INCI name	Trade Name/Supplier	% w/w
Phase A1		
Water	Water(demineralized)	Qs
Disodium EDTA	Versene Na/Dow	0.10
Glycerin	Emery 916/Cognis	2.00
Butylene Glycol	Butylene Glycol/Ruger	2.00
D Panthenol	Panthenol/BASF	0.50
Phase A2		
Xanthan Gum	Keltrol® CG-T/CP Kelco	0.10
Acrylates/C10-30 Alkyl Acrylate Crosspolymer	Carbopol Ultrez 21/Lubrizol	0.30
Sucrose Stearate	Systema SP50C /MMP	1.50
Phase B		
Simmondsia Chinensis (Jojoba Oil)	Jojoba Oil/Jeen Intl.	3.00
Shea Butter	Shebu Refined / Rita	1.50
Behenyl Alcohol	Lanette 22/BASF	1.25
Potassium Cetyl Phosphate	Amphisol® K /DSM	1.00
Stearic Acid	Pristerene® 4911/Croda	1.00
Dimethicone	Dow Corning 200,50 cst/Dow Corning	5.00
Natural Beeswax	Beeswax/Rita	1.50
Isosorbide Dicaprylate	Synovea® DOI/Sytheon	2.00
Alpha Tocopherol	Vitamin E/Basf	0.20
Ammonium Acryloyldimethyltaurate/VP Copolymer	Aristoflex AVC/Clariant	0.30
Phase C		
Sodium Hydroxide	Sodium Hydroxide (50%sol)	0.20
Phase D		
Bakuchiol	Sytenol® A /Sytheon	1.00
Phase E		
Phenoxyethanol, Ethylhexylglycerine	Euxyl PE 9010/Schulke	1.00
Phase F		
Fragrance	Women's FragranceK3258/Carrubba	0.20
Total		100.00

Procedure:

Combine A-1; disperse A2 one by one in A1 while stirring and heat A to 75°C. Combine ingredients of phase B and heat to 75 °C. Add phase B to A with good mixing. Homogenize mixture at moderate speed for 3-5 min while adding phase C (adjust pH to 5.5-6.0). Start cooling batch while adding phase D. At 40°C, add phase E & F; mix with propeller agitation until mixture is homogeneous.

Notes:

pH - 5.5-6.0 Viscosity – 80,000- 100,000mPas (Brookfield RVT, Spindle C, 10 rpm) at 25°C

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